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Feasibility, Experiences and Short-Term Outcomes on Challenging Behaviour and Well-Being of a Partially Online Music Intervention Pilot Study for Adults With Intellectual Disabilities

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ABSTRACT

Background: During the COVID-19 pandemic we implemented a partially online music intervention to examine the feasibility, experiences and short-term outcomes on the challenging behaviour and well-being of adults with intellectual disabilities.

Method: This mixed-methods study included 10 participants with mild or moderate intellectual disabilities who received 16 one-hour individual music sessions in 10 weeks, either face-to-face or online. Data on feasibility and experiences from participants and music workers were collected and analysed using inductive thematic analysis. Challenging behaviour and well-being were measured before and after intervention.

Results: Overall experiences were positive and concerned appreciation, positive feelings, musical abilities, attention span, relatedness and personalisation. Online experiences varied, but most participants preferred face-to-face over online sessions. After the intervention, challenging behaviour scores were better than before.

Conclusions: Engaging in a partially online music intervention is feasible for people with intellectual disabilities and seems to improve challenging behaviour. Experiences are discussed and recommendations for future online sessions are provided.

1 | Introduction

The COVID-19 pandemic had a profoundly negative impact upon the general population (Lai et al. 2020), and especially amongst vulnerable populations such as people with intellectual disabilities (Doody and Keenan 2021). People with intellectual disabilities were more likely to face social adversity and health inequalities than the general population when being treated in healthcare systems during the pandemic (Hassiotis et al. 2020), which magnified the disparities in healthcare (Baksh et al. 2021). Despite their high-risk status, there was a

deficiency in public health guidelines specific to people with intellectual disabilities (Desroches et al. 2021). People with intellectual disabilities faced significant challenges during the COVID-19 pandemic. The restrictions on visits and reduced face-to-face interactions with professionals (Rauf et al. 2021) led to isolation and loneliness (Embregts et al. 2020; Lake et al. 2021). Not only the quantity of formal support networks diminished, but also its quality (Scheffers, Moonen, and van Vugt 2021). Furthermore, due to disruptions in their home environment, daily schedules and occupational or recreational activities, it was difficult maintaining structure in

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their daily lives (Amor et al. 2021; Embregts et al. 2020; Navas et al. 2021). In addition, they perceived difficulty in understanding the COVID-19 preventive measures and adhering with public health measures, which was at least partly caused by an inadequacy in the provision of accessible COVID-19 information online and offline (Chadwick et al. 2022) and increased stress (Desroches et al. 2021; Embregts et al. 2020; Lake et al. 2021). This disruption of services, increased stress, and uncertainty about the future, contributed to an increase in psychiatric consultations (Rauf et al. 2021), increased challenging behaviour (Schuengel et al. 2020), and decreased well-being (Amor et al. 2021).

Interventions, either biological, psychotherapeutic or contextual, to decrease challenging behaviour and improve well-being have received considerable attention within the field of intellectual disabilities research (Bruinsma et al. 2020; Heyvaert, Maes, and Onghena 2010). One example of a non-pharmacological intervention is a music intervention, which has been shown to have a positive impact upon both challenging behaviour and well-being amongst people with intellectual disabilities (Andring 2015; Mino-Roy et al. 2021; Savarimuthu and Bunnell 2002). In addition, active music-based interventions (i.e., playing musical instruments) seem to provide more benefits, for example on challenging behaviour for people with dementia (Gómez-Gallego et al. 2021) and on well-being in the general population (Weinberg and Joseph 2017), than receptive techniques (i.e., listening to music). Within this scope, a meta-analysis concluded that an individual music intervention, with personalised attention and targeted support, is more effective than a group intervention (Pelletier 2004). In line with this, an individualised music intervention, based on personal musical preferences, is most effective in reducing challenging behaviour (Gerdner 2000) and stress (Sakamoto, Ando, and Tsutou 2013) for people with dementia. However, during the COVID-19 pandemic face-to-face music interventions required adjustments due to the imposed restrictions upon engaging with professionals in person (Rauf et al. 2021). Therefore, alternatives were required, such as innovative online interventions (Lunsky et al. 2022; Rauf et al. 2021). Online interventions have been successfully applied to people with intellectual disabilities who exhibit challenging behaviour (Oudshoorn et al. 2021; Sheehan and Hassiotis 2017), and have been found to be potentially beneficial within the social and behavioural domains (Moreno, Sans, and Fosch 2021) and as positively impacting upon their well-being (Chadwick et al. 2022). We argue that an online music intervention could also be beneficial for both the challenging behaviour and well-being of people with intellectual disabilities.

Online interventions with people with intellectual disabilities face several challenges, including securing free access to digital devices (Lussier-Desrochers et al. 2017), the digital literacy of this group and finding caregivers who are able to facilitate these online interventions (Chadwick et al. 2022). People with intellectual disabilities must also have the necessary cognitive and sensorimotor abilities in order to take part (Lussier-Desrochers et al. 2017; Sheehan and Hassiotis 2017). Indeed, video-conference therapy was not suitable for people with intellectual disabilities who experience problems with communication, social interaction or processing difficulties, and, therefore, only

phone therapy was deemed to be suitable (Rawlings et al. 2021). For online music interventions, there are additional specific challenges pertaining to, for example, latency and variable sound quality (Dowson et al. 2021). In light of these challenges, it is essential to collaborate with people to ascertain which approach suits them best (Keenan and Doody 2023) as one-size-fits-all solutions will not suffice. Consequently, it is plausible that blended interventions, that is, a combination of online and face-to-face intervention sessions, might be more suitable and acceptable than exclusively online sessions (Sheehan and Hassiotis 2017; Topooco et al. 2017; Wentzel et al. 2016).

Given the preference for a blended intervention, and the positive effect of music on challenging behaviour and well-being, we conducted a pilot study to evaluate a partially online music intervention. Specifically, this study assessed the feasibility, experiences and short-term outcomes on challenging behaviour and well-being of this partially online music intervention for people with intellectual disabilities. A concurrent mixed methods approach was used to evaluate the experiences of participants, caregivers and music workers and quantify aspects of its potential impact.

2 | Method

2.1 | Study Design

This pilot study started during the COVID-19 pandemic when physical visits were permitted only under certain conditions (Vereniging Gehandicaptenzorg Nederland 2020). The intervention consisted of 16 one-hour individual music sessions that took place over a 10-week period. Data collection from participants as well as from professionals was underpinned by a concurrent triangulation design, with both quantitative and qualitative data being collected at baseline and post-intervention and log reports after each music session and subsequently side-by-side compared (Creswell 2009).

2.2 | Ethical Approval

Ethical approval was provided by the Science and Ethics committee of the Faculty of Behaviour and Movement Sciences, VU Amsterdam (VCWE-2020-133). An independent medical ethics committee stated the research was not subject to the Dutch Medical Research Involving Human Subjects Act.

2.3 | Participants

The participants comprised 10 people (50% male) with a median age of 62.5 years (ranging from 53 to 72 years of age) who were living within three residential facilities for people with intellectual disabilities in the Netherlands. Eight participants had mild intellectual disabilities (IQ 50–69), whilst the remaining two had moderate intellectual disabilities (IQ 35–49). Four of the participants had one or more psychiatric diagnoses: two people were diagnosed with autism spectrum disorder, three were diagnosed with a mood disorder, and one participant was diagnosed with an anxiety disorder.

The inclusion criteria were as follows: (1) 18 years or older; (2) must have mild or moderate intellectual disabilities (IQ 36–70); (3) live with intensive assistance, care and behavioural regulation as described in the Dutch Long-term Care Act (Wet Langdurige Zorg, VG6) and (4) be mentally capable of giving informed consent. The exclusion criteria were as follows: (1) not capable of participating in a 1-h long intervention session, (2) have hearing problems that are not diminished via the use of a hearing aid or (3) have a medical condition (e.g., dementia) that would greatly impact upon their participation. Demographic information about the participants was provided by the special psychologist connected to the facility in which the participants lived.

Both the caregivers and special psychologist from the facilities determined which residents could potentially participate. Subsequently, potential participants and their legal representative (if applicable) were presented with an information letter and informed consent form providing detailed information about the study. Initially, there were 16 potential participants identified, however six of them ended up not participating for various reasons: three failed to hand in their informed consent form before the deadline, two stated they did not want to participate, whilst one participant withdrew on medical grounds. After the participants provided informed consent ($n = 10$) and, if applicable, their legal representative also did so ($n = 5$), music workers who lived near the facility were contacted and the first informal meetings between music workers, caregivers and participants were organised.

2.4 | Procedure

Initially, several care managers ($n = 14$) from the Dutch service provider Philadelphia Care Foundation were contacted about participating in the study. There were various reasons for their non-participation: the facility had no people who met the inclusion criteria ($n = 3$), COVID-19 related problems ($n = 3$), contact ceased after multiple attempts at communication ($n = 3$), or the facility had other priorities/organisational issues ($n = 2$), which resulted in three participating facilities.

2.5 | The Music Intervention

The sessions were guided by a music worker, who could be any individual who played an instrument and was capable of transferring their musical enthusiasm to someone with intellectual disabilities. More specifically, there were three music educators ('muziekagoog' in Dutch), one music therapy student, one psychology student with more than 15 years of experience as a musician and one professional caregiver with qualifications for music education. Most music workers had prior experience working with adults with intellectual disabilities, while others had experience working with people who have other cognitive disabilities (i.e., people with dementia). The sessions took place in a quiet setting, mostly the participants' apartment or another room of the facility where there would be no disturbances. During the sessions, the participants could choose from a box with a variety of instruments they had on loan during the entire intervention (e.g., hand drum, bongo, triangle, keyboard, and guitar). Each

participant chose their own preferred music genre and songs, sometimes with some help from the music worker.

Each music session started with a welcoming song (2 min), followed by a listening exercise (5 min), a warming up (5 min), active music making (35 min), cooling down (5 min) and a closing song (2 min). The music worker often personally adjusted the welcoming song to the participant and this song remained the same in each session. In the listening exercise, a music piece was introduced in which the participants were instructed to focus and answer questions based on that music piece. This music piece was different in each session and was played by the music worker. An alternative was for them to put on some music. Warming up exercises for both body and voice were included, tailored to match the subsequent components of actively making music: mostly instrumental performance afterwards, then the warming-up focused on movement, whereas when singing was the main component afterwards, then the warming-up included vocal exercises. In the active music making part, participants played on their preferred instruments and/or sang with the music worker. This was followed by a cooling down, in which the amount of beats per second was reduced. The music session ended with a closing song chosen by the participant. Each session was closed by the same song. The participants received a diploma after completing the final session.

The music workers and participants decided prior to each session whether the session would take place face-to-face or online, albeit under the condition that some of the sessions (no exact number was mandated) needed to take place online, in order to be able to compare the experiences of face-to-face to online music sessions. In those instances when the session was held online, a secure virtual environment was organised and the assistance of a caregiver from the facility was required. During online sessions the participant had access to all musical instruments.

The above mentioned structure and guidelines were provided in a manual with several example exercises for the music worker. This manual was prepared by the authors of the study in order to ensure consistency between the different sessions and participants (available upon request from the corresponding author). No guidelines were provided on the amount of decibel levels nor on who should be in control of the volume.

2.6 | Outcome Measures

Three non-standardised measures were used to evaluate both the feasibility of and experiences with the music intervention. Two standardised measures were used to measure challenging behaviour and well-being, both at baseline and post-intervention. Given the restrictive measures in operation during COVID-19, all questionnaires and reports were sent and received digitally.

2.6.1 | Non-Standardised Measures

2.6.1.1 | Log Report by the Music Workers. After each session, every music worker completed a log report containing a series of questions about each session: whether the session was face-to-face or online, whether the session was carried

out in accordance with the manual, which instruments were used, how much time was used for active music making, as well as if they had any other additional remarks they wished to make.

2.6.1.2 | Personal Evaluation Questionnaires. Both the participants and music workers were asked to complete an evaluation questionnaire after the final music session. This questionnaire sought to collect data on the perceived benefits, barriers and intended musical activities after the intervention period as well as their respective experiences with the face-to-face and online music sessions. The music workers also answered questions about any potential behavioural or emotional changes that they had observed in the participants over the intervention period.

2.6.1.3 | Final Meeting Questionnaire. After the final music session, the music worker, caregiver, and participant had a final meeting in which they evaluated the music sessions together with the help of a questionnaire. This helped them to discuss how the overall experiences with the music intervention were. Questions focused on what the participant, music worker and caregiver liked or disliked about the music intervention, perceived differences between the face-to-face and online sessions, possible observed changes in the participants' behaviour or emotions during the intervention and ideas for future improvement. The music worker noted down their joint answers.

2.6.2 | Standardised Measures

2.6.2.1 | The Aberrant Behaviour Checklist (ABC). The ABC (Aman et al. 1985) assesses the challenging behaviour of people with intellectual disabilities against 58 items rated on a 4-point Likert scale, with 0 being 'not at all a problem' and 3 being 'the problem is severe in degree'. It comprises five subscales: (1) Irritability, Agitation, Crying (15 items); (2) Lethargy/Social Withdrawal (16 items); (3) Stereotypic Behaviour (7 items); (4) Hyperactivity/Noncompliance (16 items) and (5) Inappropriate Speech (4 items). Higher scores indicate more challenging behaviour, with the maximum score being 174. The ABC has been found to be an adequately reliable and valid tool (Paclawskyj et al. 1997). Within the present study, the ABC was rated by caregivers of the participants. Although we stressed the importance of inter-observer reliability, unfortunately in 30% of the cases the post-intervention ABC was completed by a different caregiver than the one who completed the baseline ABC.

2.6.2.2 | The Personal Wellbeing Index-Intellectual Disability (PWI-ID). The PWI-ID measures satisfaction within seven life dimensions (Cummins and Lau 2005). Normally, there is a pre-testing protocol to determine both if and to what level of complexity scale a participant is able to understand the PWI-ID. However, we skipped this pre-testing protocol in order to reduce the test administering burden upon the participants. Therefore, we used the reduced PWI-ID version with three choices for the participants which use drawings of three faces (sad, neutral, and happy). As indicated in the manual (Cummins and Lau 2005), we rescaled the items to scores

from 0 to 100, to facilitate interpretation of the results, in which 100 is the most optimal overall well-being score. It has been found to be a reliable and valid tool for measuring the subjective well-being of people with intellectual Disabilities (McGillivray et al. 2009).

2.7 | Data Analysis

Qualitative data were analysed with inductive thematic analysis (Braun and Clarke 2006). For the first three participants, the text sections from all of the non-standardised measures that were relevant to answering the research questions, were identified and coded independently by two researchers. This was performed in steps to allow the two researchers (GS and KV or ES) to discuss the codes. In the event of doubt, the third researcher was consulted. Consensus about the codebook was reached after coding the information of the third participant, after which the first researcher (GS) coded the information of the last seven participants. All codes were grouped into themes (GS) and interpretations were jointly discussed and refined by the three authors (GS, KV and ES) in order to ensure that the codes within the themes were coherent and that the codes between the themes were distinctive.

The quantitative data from the log reports, personal evaluation questionnaires, and final meeting questionnaires were analysed via descriptive statistics. For the two standardised measures (i.e., ABC and PWI-ID) the descriptive statistics were reported. Due to the nature of this feasibility study, it was decided not to conduct any efficacy statistical tests on this data. In the event that there were two answers noted on the ABC or PWI-ID, the intermediate answer was calculated. This occurred five times for the ABC and two times for the PWI-ID. Two participants had one missing value on the PWI-ID in which case a total score was calculated by adding up the domain scores and dividing them by 6 (rather than 7). Three participants had one missing value on the ABC, which was subsequently replaced with the mean of the non-missing values on the subscale.

To ensure internal validity of the data, the researchers used data triangulation by combining multiple data sources (i.e., informants) and methods (i.e., qualitative and quantitative) (Carter et al. 2014; Creswell 2009).

3 | Results

The results are described in the order of feasibility, experiences (first overall and subsequently focused on online music sessions) and short-term outcomes. Verbatim quotes are used as typical examples and are labelled using their unique identifier.

3.1 | Feasibility

Recruitment started in July 2020 with a consent rate (participants/approached) of 62.5 and data collection ended in June 2021. Two participants did not take part in any online sessions and one participant only took part in online sessions. If we leave these three participants out, then the average number of online

sessions was 4.4 (range 1–13). The reasons that were given for not conducting (more) online sessions were the illness of an assisting caregiver, the workload of the caregiver who needed to be available to assist, problems with one participant's sight (cataract), the continuously changing work schedule of one of the participants and technical issues. The reason for not conducting any face-to-face music sessions with one participant was the choice of the music worker who had not yet received a COVID-19 vaccination at the time of the music sessions. Eight participants received all the sessions within the 10-week period, one participant needed a few days more, whilst one participant required a 14-weeklong intervention due to them often cancelling sessions at late notice.

3.2 | Experiences

After analysing the data, six themes emerged regarding experiences of the combined face-to-face and online music intervention: appreciation, positive feelings, enhanced musical ability, attention span, sense of relatedness, and finally, personalisation.

3.2.1 | Appreciation

All of the participants ($n=10$) stated they would recommend the music sessions to others and everyone reported that they would like to continue making music. Eighty percent of the participants indicated that the music sessions meant a lot to them.

3.2.2 | Positive Feelings

For all the participants, pleasure was mentioned the most by both participants and music workers. Both the participants and music workers reported that they really enjoyed making music, that they had a lot of fun during the sessions, were very enthusiastic about the music sessions and really looked forward to them. According to the music workers, the participants opened up, got goosebumps, got carried away by the music and were able to put their thoughts or feelings into the music, because some participants cried or nearly cried during sessions. Most music workers indicated that the music soothed and relaxed the participants and that the participants became surer of themselves during the music sessions. Some participants reported feeling a sense of pride, especially after receiving their diploma. In addition, caregivers mentioned that the participants showed a more pleasant and active participation in the group directly after having the music sessions. The caregivers indicated that they saw that the participants were happy and joyful, talking constantly about the music sessions and singing songs.

The enjoyment increased with every session. [Name of participant] can really enjoy music and doing it himself without restrictions and inhibitions really gave him wings. You could really see him shine (music worker 5).

[Name of supervisor] noticed that when she brought [name of participant] his medicines in the morning, he was already looking forward to the music session scheduled later that day (music worker 5).

3.2.3 | Enhanced Musical Ability

Various music workers stated that the participants' musical abilities improved over the duration of the intervention, namely in terms of their rhythm, harmony, musical scales when playing on the piano and progress with singing.

Then I explored the things he really liked in greater depth, and this greatly improved his sense of rhythm and harmony. He now plays and sings almost everything in the right time and key (music worker 5).

3.2.4 | Attention Span

Several music workers reported on the attention span and focus of the participants: (1) one participant had a very short attention span (around 10 s), (2) one participant actually fell asleep during the music session, (3) 1 h proved to be too long for one participant and she needed a pause in between, (4) one participant was easily distracted and (5) the other participants were also visibly tired at the end of the music session but were able to stay focused for the entire hour.

3.2.5 | Sense of Relatedness

An overwhelming majority of music workers felt that the contact between the music worker and participant was very important. Both music workers and participants commented on the collaboration, positive vibe, trust, and warm bond between them. They both felt a sense of connection over the course of the sessions. According to music workers, an important indicator of this connection was by making more and more eye contact with one another over the course of the sessions.

She said she thought the contact was worthwhile and might want to do more with music. She said she saw me as someone she could confide in (which I was really pleased to hear) and always liked having a chance to say what was on her mind (music worker 4).

Initially, I had to get used to the music worker, but after some time, I had created a trusting relationship and looked forward to it every time (participant 7).

3.2.6 | Personalisation

The importance of personalising the sessions to the individual wishes and needs of the participant, such as adjusting the

tempo, was a major theme according to music workers. The music workers emphasised the need to both dig deeper into the things the participants mentioned and allow space for their emotions or daily struggles to come to the fore.

He sometimes talked about his emotions, but then said we should move on to another song, and so I did (music worker 2).

3.3 | Additional Focus on Experiences With Online Sessions

Only one participant (10%) viewed the online sessions as pleasant, whereas 40% found them unpleasant or very unpleasant and 50% were either neutral or non-applicable. If there were only online sessions, then only 33% of the participants said they would participate again, whilst 44% said they did not know and 22% stated they would not participate. According to music workers, the online music sessions were unpleasant or very unpleasant for 78% of the participants. For themselves, 44% found the online music sessions to be unpleasant, 33% scored them as neutral or neutral/not pleasant, and 22% reported finding them pleasant (those who held the most online sessions).

The qualitative data analysis revealed four themes which will be discussed in turn below: negative online experiences, positive online experiences, comparison between the online and face-to-face sessions, and finally, conditions for being able to take part in the online music sessions and how to improve them.

3.3.1 | Negative Online Experiences

The most important negative online experiences were the bad connection, impersonal contact, problems with synchronicity and sound and adjusting musical methods, which will each be described below.

3.3.1.1 | Music Workers. The most reported negative online experience was the bad connection. In some instances this had to do with the specific program that was being used to enable a secure video call, which led to either insecurity about whether it would be possible to get in touch with the participant, or not being able to begin the session at the specified time. This, in turn, could sometimes lead to mutiny from the participants. The music workers expressed that the staff also often could not help with these issues. At other points, the connection was suddenly lost and participants had to either join again independently or go get help from a caregiver to do this if they were unable to do it themselves. One music worker argued that this was not helpful for the self-esteem of the participants. Another recurring point in the dataset pertained to the contact between the music workers and participants. Most of the music workers reported finding the online music session impersonal, namely because it was harder to understand the other person, harder to clarify what the participant had to do and harder to adjust to the participant. Because a lot of attention needed to be paid to the technical aspects of online sessions, it was difficult to also observe the participant. They reported being able to see only a

limited part of what was going on and, hence, that they missed the bigger picture. This, in turn, made it harder for them to react to attitudes and expressions. It was also reported that there was a lack of a sense of being together in the online sessions, which made it harder to build a bond or trust. The majority of music workers also experienced issues with synchronicity. There was a delay in the sound, making it difficult to play or sing together. Regarding the sound, in some instances the sound was too soft, whilst at other points there was reverb. Overall, there were many technical concerns, which underscores the fact that the online sessions required specific insight and technological know-how, both from the music workers and the participants.

As a music worker, you are also less able to respond to the situation if you have online sessions. One time, [name participant] had to go to the toilet for example and it took her ages to get back to the computer. At a moment like that, you have no way of seeing what's happening and I couldn't go and get her. Fortunately, she came back in the end and carried on singing enthusiastically (music worker 1).

Then we sang some more songs, along with the YouTube clips. This was quite tricky, as the client has trouble finding the songs with lyrics on YouTube by herself. When we have a face-to-face session, I normally do that for her. As it was a lot of hassle, we ended up singing 'la-la-la' to another song (music worker 1).

It was reported that musical methods routinely had to be adjusted, that in some instances it proved not to be possible to compensate for the problems the participants experienced. Making music together proved to be hard, as did explaining the exercises to the participants or showing them the right keys on a keyboard, for example. Ultimately, it was found that repeating known exercises (from the face-to-face sessions) proved to be the most effective method for some of the participants. Some music workers stated that it was exhausting and annoying to stare at a screen for long periods of time, both for themselves and participants, which in some instances led to sessions ending prematurely.

3.3.1.2 | Participants. The participants also found the online sessions to be impersonal, or as one participant referred to it: 'not real'. Some of the participants stated being more easily distracted and that their concentration was worse in these sessions. The online music sessions were a 'hassle', as one participant put it.

I can't concentrate. Very crazy, with her being on the other side (participant 3).

3.3.2 | Positive Online Experiences

3.3.2.1 | Music Workers. Several music workers stated that participants were cooperative during the online sessions.

Two music workers explained that the online sessions were an easy and fast way to reach a person, or that, as one of them stated, the fact that the online music sessions could take place in the evening or during the weekend, meant that it better fitted their own attention span. The latter of these music workers also reported experiencing a greater sense of equivalency with the participant, because they were also able to see her own living room. Given the restrictive measures in place at that juncture, one music worker commented that the online music sessions helped their participant to not feel so isolated.

Even so, I recently noticed that [name participant] really does his best. He takes the instruments out of the box when I ask him to during the online session, and sometimes plays along wholeheartedly with the music we play together (music worker 2).

3.3.2.2 | Participants. Only one participant was positive about the online sessions. The online sessions were a welcoming distraction and gave this person energy.

3.3.3 | Comparison Between Online and Face-To-Face Sessions

3.3.3.1 | Music Workers. The music workers reported that it did not feel like they were making music together with the participants when the sessions were online. They also felt that they had to work harder during the online sessions, and that participants exhibited greater agitation, showed less initiative and engaged in less eye contact. Conversely, the face-to-face sessions were experienced as being more personal, and as making it easier to explain and engage in exercises together. However, one music worker also reported that it was just as easy (or hard) to make contact with the participant during the online sessions as it was face-to-face.

3.3.3.2 | Participants. The participants answered the questions: ‘What did you think of the face-to-face music sessions?’ and ‘What did you think of the online music sessions?’ on a 5-point Likert scale. Enjoyment levels were higher for the face-to-face music sessions than they were for the online music sessions. The participants experienced the online sessions as being more impersonal and more rigid.

3.3.4 | Conditions for Taking Part in the Online Music Sessions and How to Improve Them

3.3.4.1 | Music Workers. Several conditions were mentioned by music workers with regard to being able to participate in the online music sessions which were related with the participants, technique or procedure. Regarding participants, it was mentioned that when a participant has bad sight, online sessions are practically impossible. Furthermore, online music sessions were more appropriate for participants who were capable of handling the potential stress of a bad connection. Regarding technique, several music workers noted that a system for secure video calls that works easily and that can be operated by everyone (i.e., music workers,

caregivers, and participants) is necessary. They added that providing clear instructions to everyone involved would help to make this work in a more straightforward way. Another improvement that was suggested by the music workers was to use a system that would allow participants to turn on a musical fragment themselves. The final technical point mentioned was the necessity of a computer with a stable internet connection as well as a fully charged mobile phone in some instances. With regards to procedures, music workers recommended to develop a bond and sense of trust between music workers and participants prior to starting the online sessions. In addition, they suggested that more time should be reserved for engaging in regular conversation in the programme, and that the intervention should start with face-to-face sessions for several reasons. The first reasons are to help participants understand the structure of each session, familiarise themselves with the available music instruments, and enable them to repeat exercises they already know. The final reason is for the music worker to get familiar with the living situation of their participants, for example, to help the participants navigate to plug the keyboard in the wall outlet.

3.3.4.2 | Participants. One participant reported that there needs to be a stable internet connection.

3.4 | Short-Term Outcomes on Challenging Behaviour and Well-Being

Descriptive statistics for the outcomes on the ABC and the PWI-ID are outlined in Table 1. Improvements were noted on almost all subscales from the ABC and on the total ABC score.

4 | Discussion

This pilot study evaluated a partially online music intervention for people with intellectual disabilities. Our findings suggest that such an intervention is feasible and acceptable and can improve challenging behaviour. However, the intervention protocol, trial design and one of the outcome measures will need refinements.

4.1 | Feasibility

The findings from our study show that the intervention we implemented is indeed feasible. The consent rate seemed rather low, which is not uncommon in the intellectual disabilities research field (range 37–100) (Cleaver and Sakar 2010). Not all participants participated regularly in online sessions. This finding was due to various factors, for example, participants' bad sight, illness/workload of caregivers and technical issues and one participant only had face-to-face music sessions because the music worker was not yet vaccinated.

4.2 | Experiences

Participants appreciated the partially online music intervention. Both the participants and music workers reported

TABLE 1 | Demographic information of the participant and scores on the ABC and PWI-ID.

Participant	Age range	Level of intellectual disability	ABC ^b												PWI-ID ^c	
			Total		Irritability		Lethargy		Stereotypy		Hyperactivity		Inapp. speech		Total	
			Base	Post	Base	Post	Base	Post	Base	Post	Base	Post	Base	Post	Base	Post
1	50–60	Mild	21.3	15.4	8.0	7.0	5.3	2.0	0.0	0.0	8.0	6.4	0.0	0.0	64.3	50.0
2	60–70	Mild	31.0	26.0	15.0	11.0	3.0	3.0	0.0	0.0	11.0	10.0	2.0	2.0	78.6	85.7
3 ^a	70–80	Mild	20.0	11.0	6.0	3.0	5.0	2.0	1.0	0.0	3.0	2.0	5.0	4.0	92.9	83.5
4	50–60	Mild	13.3	6.0	6.0	3.0	3.0	1.0	0.0	0.0	4.3	2.0	0.0	0.0	85.7	100.0
5	50–60	Mild	8.0	9.5	3.0	4.5	3.0	2.0	0.0	0.0	2.0	3.0	0.0	0.0	100.0	78.6
6	50–60	Mild	22.0	13.5	4.0	3.0	11.0	6.5	2.0	1.0	4.0	3.0	1.0	0.0	35.7	53.6
7 ^a	60–70	Moderate	27.0	26.0	13.0	11.0	9.0	6.5	3.0	1.0	2.0	10.0	0.0	3.0	85.7	50.0
8 ^a	60–70	Moderate	2.0	5.0	0.0	0.0	1.0	5.0	0.0	0.0	1.0	0.0	0.0	0.0	78.6	100.0
9	60–70	Mild	17.0	7.0	1.0	1.0	13.0	6.0	0.0	0.0	1.0	0.0	2.0	0.0	78.6	92.9
10	70–80	Mild	8.0	6.0	4.0	0.0	0.0	0.0	0.0	1.0	4.0	3.0	0.0	2.0	92.9	100.0
Total (mean)			17.0	12.5	6.0	4.4	5.3	3.4	0.6	0.3	4.0	3.9	1.0	1.1	79.3	79.4

Abbreviations: ABC = Aberrant Behaviour Checklist, base = baseline score, inapp. = inappropriate, post = post-intervention score, PWI-ID = Personal Wellbeing Index-Intellectual Disabilities.
^a ABC was filled in by a different caregiver post-intervention than baseline.
^b Higher scores indicate more challenging behaviour.
^c Higher scores indicate a more optimal overall well-being score.

positive, joyful experiences with the music sessions, and the overall satisfaction levels were high. Indeed, the participants really looked forward to the music sessions. In addition, the caregivers noted that participants showed a more pleasant and active participation in the group directly after their sessions. It seemed that the participants' musical abilities improved. The observations provided by several music workers highlight a range of attention span and focus levels amongst the participants during the music sessions. However, most of the participants were able to maintain focus for the duration of the hour-long sessions. The feedback from both music workers and participants underscores the relevance of the interpersonal connection between them during the sessions. This sense of relatedness that developed over the sessions was evident, with increasing eye contact cited as a key indicator of this connection. In addition, personalisation emerged as a central theme, with an emphasis on tailoring the sessions to the individual preferences and needs of each participant, which is in line with previous research (Van Der Geer et al. 2009; Van Schijndel-Speet et al. 2014).

4.3 | Additional Focus on Experiences With Online Sessions

Overall, the face-to-face music sessions were preferred to online sessions, which is in line with results from other studies, including with people with intellectual disabilities (Rawlings et al. 2021; St. John et al. 2022). Both the music workers and the participants indicated that it was very important to establish a connection with each other. Starting with face-to-face sessions before scheduling online sessions was experienced as being helpful in this regard. The importance of a prior face-to-face meeting first or working together once a connection is established has also been confirmed in other research (Lunsky et al. 2021), which makes it simpler to adapt interventions from face-to-face to online delivery (Rawlings et al. 2021). Most music workers and participants found the online sessions to be impersonal and that there was a lack of a togetherness. According to the music workers, the online sessions were more intense and tiring, both for themselves and the participants, than the face-to-face sessions. In light of this, they recommended shortening the online sessions. Other challenges with online sessions were also identified, including bad internet connections, problems with synchronicity and sound and having to adjust musical methods. Based on these challenges, it is recommended to build capacity and technical know-how regarding the use of straightforward and easy-to-use digital systems amongst care personnel, as suggested by Dowson et al. (2021). It is also important to check whether participants have a medical condition which might impinge upon their ability to engage online (as in the case of the participant with cataracts in this study). In light of these overall negative online music experiences, it is important to consider that our sample consisted predominantly of older individuals. Younger people with intellectual disabilities have, in general, more experiences with technology (Näslund and Gardelli 2013), which may lead to different findings. Positive experiences of online sessions centred on the fact that they are efficient time-wise and provide a convenient way to reach people with intellectual disabilities, which was also found by Lunsky et al. (2021).

4.4 | Short-Term Outcomes on Challenging Behaviour and Well-Being

The finding regarding the decrease in challenging behaviour after the music intervention is in accordance with previous research amongst adults with intellectual disabilities (Savarimuthu and Bunnell 2002), children with intellectual disabilities (Kalgotra and Warwal 2017), and people with dementia (Han et al. 2010; Pedersen et al. 2017). Moreover, this decrease in challenging behaviour was also apparent amongst participants who received relatively more online music sessions, thus lending further support to the idea that interventions using digital devices can be used to improve challenging behaviour (Moreno, Sans, and Fosch 2021). This pilot study was unable to corroborate the results of previous studies concerning music interventions improving well-being, including amongst people with intellectual disabilities (Savarimuthu and Bunnell 2002), people with mental illness (Chang et al. 2018), people with schizophrenia (Geretsegger et al. 2017), and those with dementia (Moreno-Morales et al. 2020; Van Der Steen et al. 2017). There are several potential explanations for this result. First, well-being scores were already high at baseline, which is in accordance with both Jaschke and Scherder's (2015) results and the 'Theory of Subjective Well-being Homeostasis' (Cummins 2005). Second, the three-point differential scale version of the reduced PWI-ID overly limited the intra-individual variability (Cummins and Lau 2005), whilst the 10-week period might have also been too short to improve well-being. Nevertheless, it did appear that the participants maintained their well-being despite the fact that the COVID-19 pandemic was worsening during that time (the increasing number of infected persons had led to more restrictions). Whether or not this intervention contributed to the maintenance of participants' well-being despite the tumultuous circumstances at that time, as is suggested in other research (St. John et al. 2022), remains unclear. Future research with a control group is recommended to explore this further.

Although this study placed more emphasis on the qualitative findings, converging the qualitative and quantitative data leads to a more comprehensive analysis of the research questions (Creswell 2009). Comparable findings are visible regarding challenging behaviour. Information from the log reports, evaluation questionnaires and final meeting questionnaire showed that the music sessions soothed and relaxed the participants and resulted in more pleasant behaviour in the group directly afterwards. In addition, participants were found to be happy and joyful and felt a sense of connection with the music worker. However, an expected improvement in well-being was not demonstrated and was likely impacted by the aforementioned issues with the PWI-ID and the measurement of well-being. Based on the predominantly negative experiences with the online music sessions, it is crucial to address the implications of these findings. While the sessions overall appeared to have a positive impact on challenging behaviour, as evidenced by both qualitative and quantitative data, and is positively associated with well-being, as evidenced by the qualitative data, most participants did not enjoy the online music sessions. Although caregivers perceived benefits from the intervention (i.e., reduction of challenging behaviour), it is essential to critically evaluate which aspects of the sessions

truly benefitted the participants and are worthwhile for them. It is evident that participant enjoyment and engagement are vital factors to consider in designing and implementing interventions tailored to this population. Reducing the number of online sessions to a minimum for those who do not prefer online sessions could be a viable approach to better align with participants' preferences and needs.

There are a number of limitations in this pilot study. Besides the aforementioned limitations of the reduced PWI-ID and lack of a control group, this study was also limited by its small number of participants and their relatively older age, which limited the generalisation of the findings to all adults with intellectual disabilities. Furthermore, there might be selection bias in our sample because the participants were interested in a music intervention and willing to perform the music sessions, further limiting the generalisation of the results. In addition, the use of a self-constructed questionnaire to gather participant and music worker experiences, presents additional considerations regarding the reliability and validity of our findings. Furthermore, the reliance on self-report measures introduces the possibility of response bias, whereby participants and music workers may provide more favourable or socially desirable responses. In our study, the ABC introduced another additional challenge for potential inter-rater bias, because for three participants, the ABC at post-intervention was scored by another person than at baseline. Despite this difference in raters, the difference in scores for these three participants seemed consistent with the scores from the other participants. Further, one music worker and three participants only experienced one type of session (i.e., face-to-face or online), which limited their feedback only to that type of session. Lastly, we did not collect data on the genre of music that participants or music workers chose during the music intervention nor on the music worker's skillset, which both could have impacted the findings. Music genre preferences are associated with emotional regulation strategies (Cook, Roy, and Welker 2019). Furthermore, given the variability in music workers' experiences and education, future studies should consider using observational measures. This would help investigate the music workers' approach, attitude and skills, which is important for the successful implementation of music interventions (Savarimuthu and Bunnell 2002) and could shape participants' experiences. Given these limitations, the findings of this study should be interpreted with caution and it calls for replication studies on a wider scale with a more diverse sample.

With regards to future research, the findings of this study suggest some necessary modifications in order to tailor an online music intervention appropriately. For example, it would be valuable to investigate the experiences and effects of a (partially) digital music intervention that starts with face-to-face sessions to build up a relationship first, incorporates shorter online sessions and employs easier digital systems that deal with the problem of synchronicity. These modifications could potentially enhance participant enjoyment and clinical practice. Additionally, further research is needed to examine how individual characteristics interact with the format and content of online music sessions to influence engagement and outcomes. In light of the challenges posed by the COVID-19 pandemic and the negative impact it has had on people with intellectual disabilities, there is a growing recognition of the importance of establishing flexible support

options, including online interventions. However, it is crucial to establish a balance and ensure that the shift to (partially) online formats is implemented wisely, particularly for older individuals with intellectual disabilities who may face additional barriers to engaging with digital platforms. By adapting interventions based on participant feedback and preferences, we can enhance the overall quality of support provided to this population.

5 | Conclusion

Overall, this study explored the feasibility of a partially online music intervention consisting of 16 sessions over a 10-week period for adults with intellectual disabilities living in residential facilities. This initial data shows that both the participants and music workers generally reported positive experiences concerning taking part in the music intervention, and the overall satisfaction of the participants was high. The face-to-face music sessions were preferred to online music sessions. Furthermore, the challenging behaviour of these participants showed improvements, whereas the results concerning well-being were not conclusive. Balancing between making optimal use of online formats and addressing the unique needs and challenges faced by adults with intellectual disabilities is paramount. This study helped in terms of identifying common issues associated with conducting online music sessions for especially older adults with intellectual disabilities and providing recommendations for music workers to successfully adopt this type of support for people with intellectual disabilities. As such, although it was a pilot study with a small sample size, it has important implications for future research in this area given the restrictive measures due to the COVID-19 pandemic and the disparities of access to interventions by people with intellectual disabilities.

Author Contributions

Gerianne J. J. Smeets: conceptualization, formal analysis, writing – original draft, project administration. **Karin M. Volkers:** conceptualization, formal analysis, writing – review & editing, supervision. **Eline A. K. Swart:** formal analysis, visualisation, writing – review & editing. **Xavier M. H. Moonen:** conceptualization, writing – review & editing, supervision. **Erik J. A. Scherder:** conceptualization, writing – review & editing, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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